Abdelrahman Khalifa

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Summary

Software engineer with **14 merged upstream PRs since March 2025**: 11 in OpenPrinting (libcups, the CUPS library in C, and its Rust bindings cups-rs) as a Google Summer of Code 2026 contributor with The Linux Foundation (OpenPrinting), plus Rust Clippy, Boa and Zephyr RTOS. Every change went through maintainer code review. Comfortable building and testing C and Rust projects locally and in CI against real services, and explaining changes clearly in writing.

Experience

Google Summer of Code Contributor May 2026 – Oct 2026
The Linux Foundation (OpenPrinting) *Remote*

- Built the **Printers page for COSMIC Settings** and a standalone app sharing the same UI: printers grouped by device, status and supply levels, print queue control, test pages, live updates, and an Add Printer flow that sets up legacy printers through Printer Applications (PAPPL) instead of PPD drivers (cosmic-settings #2182, in review).
- Ran the backend as a varlink IPC service in `cosmic-settings-daemon` (cosmic-settings-daemon #188, in review), so the C FFI code never loads in the Settings process.
- Got **11 PRs merged upstream** in OpenPrinting (8 during the program). cups-rs (Rust): ported it to libcups3 with CUPS 2 and CUPS 3 builds and a CI matrix across architectures; added IPP management operations, Printer Application support, IP/port connections, `lpoptions` and FFI bindings for the libcups3 DNS-SD API (9 PRs). libcups (C): fixed crashes in `cupsDNSSDDelete()` and a leaked client in the Avahi DNS-SD code (#164, #165).

Embedded Software Engineer Aug 2025 – Present
EKSON *Egypt*

- Wrote the ESP32-S3 firmware and migrated it from ESP-IDF/FreeRTOS to Zephyr RTOS: PID valve control with IMU feedback, safety stops and a UART protocol.
- Developed the PyQt5 desktop control app, cutting CPU usage and latency by about 40% (measured) using asyncio to wait for ESP32 and server messages concurrently.

Embedded Linux Systems Intern Dec 2024 – Jan 2025
Orange Digital Center (ODC) *Egypt*

- Created custom Linux images with Yocto and Buildroot and wrote a Linux kernel module; worked with device trees, root file systems and hardware flashing. Scored 100% in the final evaluation.

Projects

Kernel Monitor | *C, Linux, ARM, QEMU* GitHub

- Character device driver that walks the process list under RCU and exports per-task IDs, memory, threads, context switches and CPU time to user space; run on QEMU.

Database Systems, CMU 15-445 | *C++17* GitHub

- Thread-safe Count-Min Sketch for the BusTub database (one mutex per row; insert, count, merge, top-k) and SQL analytics in DuckDB.

Technical Skills

Languages: Rust, C, Modern C++, Python (asyncio), Bash, JavaScript (Node.js)
Systems & Linux: Linux (Ubuntu/Debian), operating systems, FFI and bindings, IPC (varlink, D-Bus; zbus, zlink), concurrency (async/event-driven, multithreading), memory management, networking (TCP/UDP), CUPS/IPP, DNS-SD/mDNS (Avahi), kernel modules and device drivers
Tooling: Code review, testing, Git/GitHub, GitHub Actions CI (multi-architecture matrices), Cargo, Clippy, rustfmt, CMake, Make, Yocto, Buildroot, QEMU, debugging, performance optimization

Education

Minia University Minia, Egypt
Bachelor of Science (B.Sc.), Computer and Systems Engineering *2020 – 2025*

- Grade: Very Good with Honors.
- Activities: Codeforces Expert; 4-time ECPC finalist (2021–2024); Treasurer, IEEE Student Branch.